

CHENGLONG YU

(+86) 198-6221-6761 · yuchenglong756@gmail.com · chenglongyu.com

EDUCATION

Institute of Computing Technology, Chinese Academy of Sciences (ICT, CAS) Beijing, China
Master of Engineering in Computer Technology Sep. 2024 – Expected Jun. 2027
China University of Petroleum (East China) Qingdao, China
Bachelor of Engineering in Computer Science and Technology Sep. 2020 – Jun. 2024
GPA: 4.12/5 | Rank: 3/114 (Top 3%)

RESEARCH INTEREST

Reinforcement Learning, Knowledge Distillation, Diffusion Models, World Models, Resource Allocation

RESEARCH EXPERIENCE

Joint Task Offloading and Resource Allocation with Multi-Task Energy Control based on World Model-Guided Collaborative Multi-Agent Deep Reinforcement Learning

Independent Research Dec. 2023 - Nov. 2025

- **Proposed** a multi-task energy control constrained joint optimization scheme for the task offloading and resource allocation (MEC-JTORA) to maximize the terminal energy efficiency.
- **Designed** an error-corrected energy consumption model (ECECM) to allocate appropriate energy to tasks.
- **Developed** a World Model-guided Collaborative Multi-Agent Deep Reinforcement Learning (WM-MADRL) algorithm, leveraging the world model to accelerate convergence and guide policy exploration.
- **Achieved** a 36%~89% improvement in convergence speed over baseline algorithms, and increased the total size of completed tasks and energy efficiency by 4~9 times compared to systems with equal energy allocation.

Sample-Efficient Data-Free Knowledge Distillation

Independent Research Aug. 2025 - Present

- **Proposed** a diversity- and difficulty-guided sample synthesis method to tackle the inefficiency of current Data-Free Knowledge Distillation (DFKD).
- **Designed** a batch-based feature repulsion mechanism to guarantee the intra-class diversity of synthetic samples, and introduced a sub-optimal class boosting strategy to ensure sample difficulty.
- **Integrated** the proposed method into the Fast-DFKD (AAAI 2022), improving the accuracy of the distilled student models by 1.30%~1.83% on CIFAR-10 and 1.68%~3.95% on CIFAR-100.

PUBLICATIONS & PATENTS

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- **C.Yu**, W. Xing, J. Shi, Y. Zhou and L. Liu, "Collaborative Multi-Agent Deep Reinforcement Learning for Joint Task Offloading and Resource Allocation with Long Term Energy Control," 2025 IEEE 102nd Vehicular Technology Conference (VTC2025-Fall), Chengdu, China, 2025, pp. 1-5. [pdf]
 - **C.Yu**, Y. Zhou, Y. Qi and J. Wang, "Joint Task Offloading and Resource Allocation with Multi-Task Energy Control based on World Model Guided Collaborative Multi-Agent Deep Reinforcement Learning," IEEE Internet of Things Journal (under review).
 - Y. Zhou, **C. Yu**, J. Shi, Y. Qi, W. Xing, and L. Liu, "An Industrial Internet System for Resource Allocation Based on Collaborative Multi-Agents." Chinese National Invention Patent, Patent No.: CN202511002125.2.
 - D. Tan, **C. Yu**, Y. Zhou, L. Liu, and J. Shi, "A Data Aggregation Method and System for Wireless Ad Hoc Networks." Chinese National Invention Patent, Patent No.: CN202411514682.8.

HONORS & AWARDS

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- Outstanding Graduate of Shandong Province - 2024
 - National Encouragement Scholarship (Three times) - 2021, 2022, 2023
 - UCAS Academic Scholarship (Twice) - 2024, 2025
 - Honors Bachelor's Degree, China University of Petroleum (East China) - 2024

TECHNICAL SKILLS

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- **Technology stack:** Python, C, C++, Pytorch, Linux, Git, CUDA